

Design:

↳ Blueprint

① Data Type: a kind of data.

152 → integer. (Numbers)

"John" → String (Alphabets/Special Characters)

15.7 → Float (Decimals)

'J' → Character

'?' → Character

True/False → Boolean (Logical values)

② Data Structure:

↳ way of organising & storing data

→ Student_name1 = "John"

→ Student_name2 = "Josh"

⋮

Student_name100 = "walker"

Array: Store multiple data values having same datatype.

↳ Index: Location, number.

→ SName[0] = "John"

SName[1] = "Josh"

⋮

SName[99] = "walker"

Index always starts from 0.

Same array name but different index

Total numbers = Index + 1

Highest index = Total Number - 1

SName[99] = ["John", "Josh", ..., "walker"] //

Dimension: 1D array (Row)

2D array (Rows, Columns)

$\begin{bmatrix} 1 & ['A', 'B'] \\ 2 & ['C', 'D'] \end{bmatrix}$

$\begin{bmatrix} \begin{matrix} C_1 & C_2 \\ ['A', 'B'] \end{matrix} & \begin{matrix} C_1 & C_2 \\ ['C', 'D'] \end{matrix} & \begin{matrix} C_1 & C_2 \\ ['E', 'F'] \end{matrix} \end{bmatrix}$
Row 1 Row 2 Row 3

ArrayName[R][C]

ArrayName[2][1] = 'C'

Parallel 1D array:

Name[3] = $\begin{matrix} 0 & 1 & 2 \\ ['A', 'B', 'C', 'D'] \end{matrix}$

Marks[3] = ['52', '65', '85', '70']

Name[2] = C

Marks[2] = 85

Limitation: Maintain corresponding indexes perfectly.

Records: Groups of related fields of one entity/user.

Student Name: C.

Name[2] =

Marks[2] =

PName[2] =

HPhone[2] =

⋮

Array of records:

Student[0] = ["Name: A1, Class: 8, Grad: A"]